AnalystLab Africa Week 3 Report: SQL & Data Querying

Name: Goodness Okoro
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Focus Area: SQL & Data Querying
Datasets Used: Chinook Database and Sample Sales Dataset

Introduction

This report presents my Week 3 assignment for the AnalystLab Africa Data Analytics Internship Program. The focus of this week was SQL and data querying, using relational database concepts to extract, transform, and analyze structured data.

The goal of the assignment was to practice writing SQL queries that answer real business questions. I worked with two datasets: the Chinook Database, which represents a digital music store, and the Sample Sales Dataset, which contains sales transaction records across customers, products, countries, and order dates.

The analysis covered basic SQL concepts such as SELECT, WHERE, GROUP BY, HAVING, ORDER BY, COUNT, SUM, and AVG. It also included more advanced SQL concepts such as joins, subqueries, common table expressions, window functions, ranking, and indexing.

Dataset Overview

The first dataset used was the Chinook Database. This database represents a music store and contains tables such as Customer, Invoice, InvoiceLine, Track, Genre, Employee, Album, and Artist. The main business flow in the database is that customers make purchases, purchases are recorded as invoices, invoices contain invoice lines, and invoice lines are linked to tracks and genres.

After importing the Chinook Database into MySQL, I confirmed that it contained 59 customers, 412 invoices, and 3,503 tracks.

The second dataset used was the Sample Sales Dataset. This dataset contains order-level sales information such as order number, quantity ordered, price, sales amount, order date, product line, product code, customer name, country, deal size, and order status.

After importing the Sales Dataset into MySQL, I confirmed that it contained 2,823 order lines and 307 unique orders. I also created a cleaned sales table called sales_clean, where the order date was converted from text into a proper date format for time-based analysis.

Chinook Database Analysis

The Chinook Database analysis focused on understanding customer purchases, revenue performance, country performance, genre performance, employee performance, and customer spending behavior.

The first query provided a revenue overview of the music store. The result showed that the store recorded 412 invoices, generated a total revenue of $2,328.60, and had an average invoice value of $5.65. This suggests that most purchases in the store were small-value music transactions.

The second query analyzed revenue by billing country. The USA generated the highest revenue with $523.06 from 91 invoices. Canada followed with $303.96, while France generated $195.10. This shows that the USA was the strongest revenue market for the Chinook store.

The third query identified the top-spending customers. Helena Holý from the Czech Republic was the highest-spending customer, with a total spending of $49.62 across 7 invoices. Richard Cunningham from the USA followed with $47.62, while Luis Rojas from Chile spent $46.62. The result showed that some customers had the same number of invoices but different spending values.

The fourth query analyzed monthly revenue trends. The highest monthly revenue observed was $52.62. Some of the peak months included January 2022, April 2023, June 2023, and November 2025. This shows that revenue was not evenly distributed across all months, making time-based analysis useful for identifying sales peaks.

The fifth query analyzed music genres by revenue. Rock was the best-performing genre, generating $827 from 835 items sold. Latin followed with $382, while Metal generated $261. This shows that Rock was the strongest-performing genre in the Chinook music store.

The sixth query analyzed sales support agent performance. Jane Peacock generated the highest customer revenue among the sales support agents, with $833.04. She supported 21 customers and was linked to 146 invoices. Margaret Park followed with $775.40, while Steve Johnson generated $720.16. This suggests that Jane Peacock had the strongest sales support performance.

The seventh query used window functions to rank customers within each country based on total spending. This helped identify the highest-spending customer in each country. The query demonstrated the use of RANK(), ROW_NUMBER(), and PARTITION BY.

The eighth query used a subquery to identify customers whose total spending was above the average customer spending. The result showed that 22 customers spent above the average. Helena Holý had the highest above-average spending at $49.62. This helps identify valuable customers who may be suitable for loyalty or retention campaigns.

Sales Dataset Analysis

The Sales Dataset analysis focused on overall sales performance, product line performance, country performance, customer contribution, monthly sales trends, product-level performance, customer value, and deal status analysis.

The first query provided an overall sales summary. The dataset contained 2,823 order lines from 307 unique orders. Total sales amounted to $10,032,628.85, with an average sales value of $3,553.89 per order line. The dataset covered transactions from January 2003 to May 2005.

The second query analyzed product line performance. Classic Cars was the best-performing product line, generating $3,919,615.66 from 199 orders. Vintage Cars followed with $1,903,150.84, while Motorcycles generated $1,166,388.34. Trains generated the lowest sales at $226,243.47. This shows that Classic Cars contributed the largest share of revenue.

The third query analyzed sales by country. The USA generated the highest sales revenue with $3,627,982.83 from 112 orders. Spain followed with $1,215,686.92, while France generated $1,110,916.52. This shows that the USA was the strongest country market in the Sales Dataset.

The fourth query identified the top customers by sales. Euro Shopping Channel was the highest revenue-generating customer, with $912,294.11 from 26 orders. Mini Gifts Distributors Ltd. followed with $654,858.06 from 17 orders. Australian Collectors, Co. ranked third with $200,995.41. This suggests that a small number of customers contributed a large share of total revenue.

The fifth query analyzed monthly sales trends. November 2004 recorded the highest monthly sales, with $1,089,048.01 from 33 orders. November 2003 was the second-highest month, generating $1,029,837.66 from 28 orders. October also appeared among the top-performing months in both 2003 and 2004. This suggests that sales were strongest around the last quarter of the year, especially in October and November.

The sixth query identified the top specific products by sales. Product S18_3232 was the highest revenue-generating product, with $288,245.42 from 52 orders. S10_1949 followed with $191,073.03, while S10_4698 generated $170,401.07. Most of the top products belonged to the Classic Cars product line, supporting the earlier finding that Classic Cars was the strongest-performing category.

The seventh query used a window function to rank products within each product line. This helped identify the best-performing product inside each product category. S18_3232 ranked first in Classic Cars, S10_4698 ranked first in Motorcycles, S18_1662 ranked first in Planes, and S12_1666 ranked first in Trucks and Buses. This query showed how RANK() and PARTITION BY can be used for category-level performance analysis.

The eighth query used a subquery to identify customers whose total sales were above the average customer sales value. The result showed that 31 customers generated above-average sales. Euro Shopping Channel was the highest above-average customer, generating $912,294.11 from 26 orders. Mini Gifts Distributors Ltd. followed with $654,858.06 from 17 orders.

The ninth query analyzed sales by deal size and order status. Medium-sized shipped deals generated the highest sales, with $5,633,596.52. Small shipped deals followed with $2,477,510.26, while large shipped deals generated $1,180,394.30. Most revenue came from shipped orders, showing that completed orders contributed the largest share of sales.

6. Query Optimization

Query optimization was considered by identifying columns frequently used in joins, grouping, filtering, and ordering. For the Chinook Database, indexes were suggested on columns such as CustomerId, InvoiceId, TrackId, and GenreId because these columns are used to connect related tables.

For the Sales Dataset, indexes were suggested on columns such as Country, CustomerName, ProductLine, OrderDate_Clean, DealSize, and Status. Some imported text columns required prefix lengths when creating indexes because MySQL requires a specified key length for TEXT/BLOB columns.

In addition to indexing, the queries were written with clear aliases, readable formatting, Common Table Expressions, and appropriate filtering. This made the SQL script easier to understand, maintain, and review.

Conclusion

This Week 3 assignment strengthened my understanding of SQL as a tool for data querying and business analysis. By working with both the Chinook Database and the Sales Dataset, I practiced importing data into MySQL, understanding schema relationships, writing analytical queries, and extracting insights from structured data.

The Chinook analysis showed that the USA was the strongest revenue market, Rock was the top-performing genre, Jane Peacock was the highest-performing sales support agent, and 22 customers spent above the average customer spending level.

The Sales Dataset analysis showed that total sales amounted to $10,032,628.85, Classic Cars was the best-performing product line, the USA was the strongest country market, Euro Shopping Channel was the top customer, and November 2004 recorded the highest monthly sales.

Overall, this task helped me improve my confidence in SQL, especially in using joins, grouping, subqueries, window functions, and indexing concepts to solve business-driven analytical questions.